

Conditions that Guarantee Parallelism

Jonathan R. Bacolod

Two lines are **parallel** if they never intersect and remain the same distance apart.

Transversal

A **transversal** is a line that intersects two or more lines.

Corresponding Angles

If a transversal intersects two lines and the **corresponding angles** are congruent, then the lines are parallel.

Corresponding angles congruent



Lines are parallel

Alternate Interior Angles

If a transversal intersects two lines and the **alternate interior angles** are congruent, then the lines are parallel.

Alternate interior angles congruent



Lines are parallel

Alternate Exterior Angles

If a transversal intersects two lines and the **alternate exterior angles** are congruent, then the lines are parallel.

Alternate exterior angles congruent



Lines are parallel

Same-Side Interior Angles

If a transversal intersects two lines and the **same-side (consecutive) interior angles** are supplementary, then the lines are parallel.

Same-side interior angles sum to 180°



Lines are parallel

Examples

Example 1

Two corresponding angles measure $(3x + 10)^\circ$ and $(5x - 20)^\circ$.

For parallel lines, corresponding angles must be congruent.

$$3x + 10 = 5x - 20$$

$$x = 15$$

Example 2

Two alternate interior angles measure $(4x + 5)^\circ$ and $(6x - 35)^\circ$.

For parallel lines, alternate interior angles must be congruent.

$$4x + 5 = 6x - 35$$

$$x = 20$$

Example 3

Two same-side interior angles measure $(2x + 30)^\circ$ and $(4x + 18)^\circ$.

For parallel lines, their measures must add to 180° .

$$(2x + 30) + (4x + 18) = 180$$

$$6x + 48 = 180$$

$$x = 22$$

Example 4

Two corresponding angles measure 75° and 75° .

Since the corresponding angles are congruent,

$$75^\circ = 75^\circ.$$

Therefore, the lines are parallel.

Example 5

Two same-side interior angles measure 95° and 85° .

$$95^\circ + 85^\circ = 180^\circ.$$

Therefore, the lines are parallel.

Practice Exercises

Practice 1

Two corresponding angles measure $(3x + 10)^\circ$ and $(5x - 20)^\circ$.

Determine whether the lines described are parallel.

Practice 2

Two alternate interior angles measure $(4x + 5)^\circ$ and $(6x - 35)^\circ$.

Determine whether the lines described are parallel.

Practice 3

Two alternate exterior angles measure $(7x - 8)^\circ$ and $(5x + 20)^\circ$.

Determine whether the lines described are parallel.

Practice 4

Two same-side interior angles measure $(2x + 30)^\circ$ and $(4x + 18)^\circ$.

Determine whether the lines described are parallel.

Practice 5

Two corresponding angles measure 75° and 75° .

Determine whether the lines described are parallel.

Practice 6

Two alternate interior angles measure 110° and 110° .

Determine whether the lines described are parallel.

Practice 7

Two same-side interior angles measure 95° and 85° .

Determine whether the lines described are parallel.

Practice 8

Two alternate exterior angles measure 130° and 130° .

Determine whether the lines described are parallel.

Practice 9

$m\angle 1 = 62^\circ$ and its corresponding angle measures 62° .

Determine whether the lines described are parallel.

Practice 10

A pair of same-side interior angles measures 118° and 62° .

Determine whether the lines described are parallel.

Activities

Activity 1

Two corresponding angles measure $(2x + 18)^\circ$ and $(4x - 6)^\circ$.

Determine whether the lines described are parallel.

Activity 2

Two alternate interior angles measure $(3x + 12)^\circ$ and $(5x - 18)^\circ$.

Determine whether the lines described are parallel.

Activity 3

Two alternate exterior angles measure $(8x - 14)^\circ$ and $(6x + 10)^\circ$.

Determine whether the lines described are parallel.

Activity 4

Two same-side interior angles measure $(3x + 24)^\circ$ and $(5x + 12)^\circ$.

Determine whether the lines described are parallel.

Activity 5

Two corresponding angles measure 68° and 68° .

Determine whether the lines described are parallel.

Activity 6

Two alternate interior angles measure 124° and 124° .

Determine whether the lines described are parallel.

Activity 7

Two same-side interior angles measure 102° and 78° .

Determine whether the lines described are parallel.

Activity 8

Two alternate exterior angles measure 145° and 145° .

Determine whether the lines described are parallel.

Activity 9

$m\angle 1 = 57^\circ$ and its corresponding angle measures 57° .

Determine whether the lines described are parallel.

Activity 10

A pair of same-side interior angles measures 121° and 59° .

Determine whether the lines described are parallel.

Thank you!